RESEARCH PROTOTYPE · STATUS BRIEF

# Can an AI-assisted system tell when a CFD simulation is wrong?

socrates-cfd · OpenFOAM v2412 · benchmark: 2D flow past a cylinder at Re = 100 · 19 October 2026 · results are partial; the benchmark is still running

Simulation software can finish without a single error and still give the wrong number, because of a mesh that is too coarse, a domain that is too small, or a typo in the set-up. Today an expert catches this by running extra simulations by hand. This prototype automates that: it runs a simulation, gathers the evidence, decides whether the answer can be trusted to the requested accuracy, and runs targeted extra simulations when it cannot yet tell. An LLM may *choose* the next experiment, but it can never decide the verdict.

**6 / 7**
clearly broken cases that OpenFOAM's own diagnostics wave through

**0**
bad answers accepted by the deterministic verifier so far (static or adaptive)

**8×**
less compute when the LLM picks the experiment, on the clearest case (B1)

**±0.9 %**
uncertainty of the in-house reference drag that every case is scored against

## How it works

**1 · Simulate**
OpenFOAM runs the case, generated from a structured problem spec

→

**2 · Evidence**
Logs, mesh quality, force histories and statistics, extracted deterministically

→

**3 · Checks**
Rules with cited thresholds → PASS / FAIL / INCONCLUSIVE, plus an error budget

→

**4 · Critic**
LLM (or a fixed rule list) picks one of 4 allowed experiments

→

**5 · Verdict**
ACCEPT / REJECT / INCONCLUSIVE, always from the deterministic checks

The four experiments are: refine the mesh, shrink the time step, enlarge the domain, run longer. Each result shows how far the answer moves, and that gives an uncertainty estimate (Richardson / Celik GCI). The loop repeats until the verdict is decisive or the budget runs out (≤ 6 experiments, ≤ 12× the original compute).

**Authority limits on the LLM.**

- It sees only recorded evidence, with run names anonymised so they cannot leak which flaw a case has.
- It can only return a request for one validated experiment, and it has no shell access.
- A proposed ACCEPT or REJECT is refused and fed back as an authority violation.
- Every number it writes is audited against the evidence; untraceable numbers are counted.

## What exists today

| Component | Status |
|---|---|
| Simulation pipeline, evidence extraction, deterministic verifier, 4 experiment primitives | done 99 unit tests and 8 OpenFOAM integration tests; every run content-addressed with full provenance |
| LLM critic and autonomous loop | done blind, schema-constrained, number-audited, authority-limited |
| Benchmark: 17 cases + 3 attack cases, 6 systems compared | running about one-third complete; roughly 10 more hours on the 4-core machine |
| Baseline validation and in-house reference | done four study stages, three adversarial fact-checks |
| Independent reviews | done 45 defects found and fixed before any result was used (code, evaluation, reports) |
| Final report (13 research questions, 5 pre-registered success criteria) | to do after the benchmark finishes |

## The benchmark

The question: the time-mean drag coefficient and the shedding frequency (Strouhal number) of a 10 mm rod in a slow air stream, each to **±3 %**. The benchmark has one careful baseline, two careful variants, two borderline variants, and twelve plausible-but-wrong set-ups: a coarse mesh, a domain that is too small, a large time step, a steady solver, the wrong viscosity, the wrong force reference, premature averaging, and others. Each case is scored against the in-house converged reference (page 3).

| System | What it does |
|---|---|
| **A** | OpenFOAM's own diagnostics only: solver completed, mesh check passed |
| **B** | The deterministic verifier on the single submitted run (no extra simulations) |
| **L** | The LLM judging the evidence of the single run: no checks, no experiments |
| **C-rule** | Adaptive: verifier + experiments chosen by a fixed rule list (statistics → mesh → time → domain) |
| **C-llm** | Adaptive: verifier + experiments chosen by the LLM critic (the full system) |
| **Bexh** | Fixed exhaustive protocol: run every experiment, then verify (tests whether adaptivity matters) |

## Results so far: the seven clearly broken cases

These cases are wrong by more than 3 % whatever the reference uncertainty, or they solve the wrong problem. The correct verdict is REJECT. INCONCLUSIVE is safe but undecided. ACCEPT is a false acceptance, the dangerous outcome.

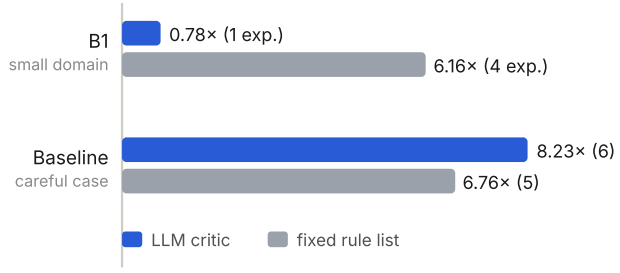
| Case | Flaw | A | B | L | C-rule | C-llm |
|---|---|---|---|---|---|---|
| A1 | mesh 2× too coarse (St −5 %) | ACCEPT × | INCONCL. | REJECT ✓ | REJECT ✓ | running |
| B1 | domain far too small (Cd +15 %) | ACCEPT × | INCONCL. | REJECT ✓ | REJECT ✓ | REJECT ✓ |
| B2 | domain too small (Cd +4 %) | ACCEPT × | INCONCL. | REJECT ✓ | running | running |
| D1 | steady solver on a periodic flow | REJECT ✓ | REJECT ✓ | REJECT ✓ | REJECT ✓ | REJECT ✓ |
| F1 | first-order convection (St −23 %) | ACCEPT × | INCONCL. | REJECT ✓ | running | running |
| H1 | dynamic viscosity entered (solves Re = 83) | ACCEPT × | REJECT ✓ | REJECT ✓ | REJECT ✓ | REJECT ✓ |
| I1 | wrong force reference values (Cd −11 %) | ACCEPT × | REJECT ✓ | REJECT ✓ | REJECT ✓ | REJECT ✓ |
| **So far** | | **6 false accepts** | 3 of 7 caught, 0 false | 7 of 7 | 5 of 5 decided | 4 of 4 decided |

### What the adaptive loop adds

On **B1** the static verifier cannot decide, and its checks all pass. The LLM critic read the evidence and asked for one experiment, enlarging the domain. Drag moved by 8.9 % and St by 6.7 %, both more than the 3 % tolerance, so the case was rejected. The fixed rule list reached the same verdict, but only after refining the mesh twice and halving the time step first.

This is the pre-registered "primary demonstration": OpenFOAM sees nothing wrong, the static verifier is inconclusive, and one targeted experiment exposes the flaw quantitatively.

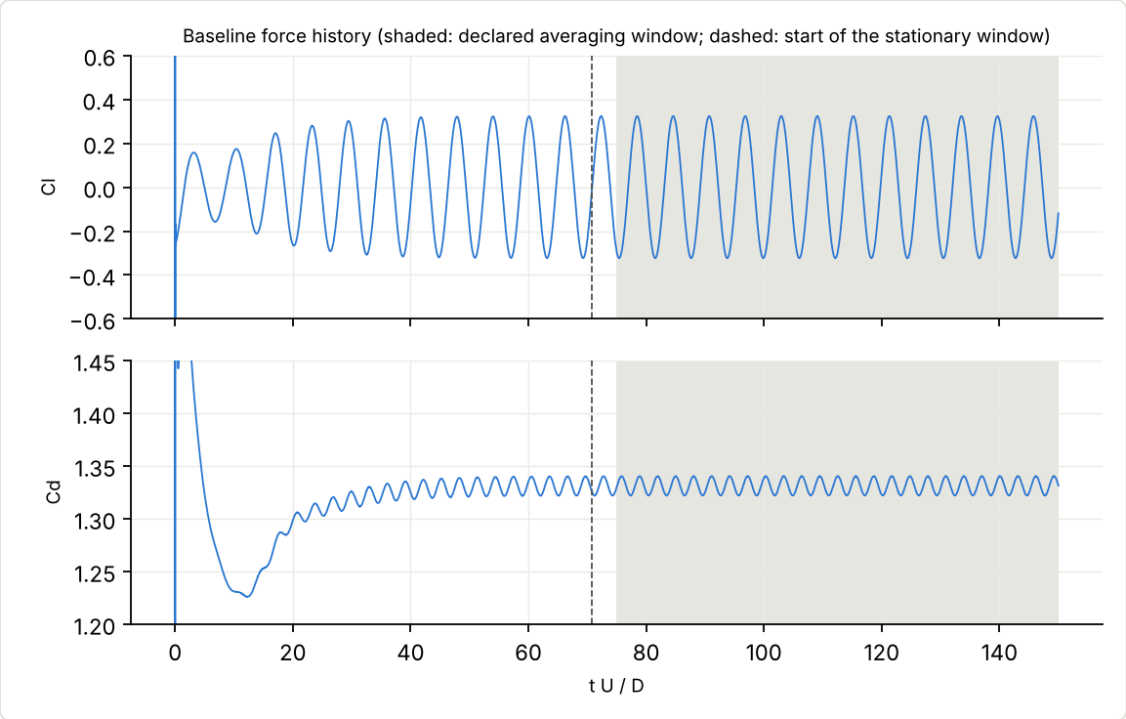
### Extra compute, relative to the original run




Both systems reached the same verdict on each case: REJECT on B1, INCONCLUSIVE on the baseline. The LLM was 8× cheaper on B1 but more expensive on the baseline (cost $0.60 in LLM calls), so whether the LLM adds value overall is not settled until the benchmark finishes.

## The baseline and the reference answer

Scoring every case needs a "true" answer, and published values are only good to about 1 % because they come from finite domains and different codes. So the project built its own converged reference. It runs the baseline at finer meshes, smaller time steps and larger domains, and extrapolates each effect to zero (Richardson extrapolation).




Baseline lift (top) and drag (bottom) over 150 flow-through times. Shedding settles after about 70; the shaded window is the declared averaging window.

| | Drag Cd | Strouhal St |
|---|---|---|
| Baseline result | 1.3313 | 0.1634 |
| **Reference** (unbounded, converged) | **1.3155 ± 0.94 %** | **0.1639 ± 0.46 %** |
| Baseline error | +1.20 % | −0.31 % |
| Literature band (finite domains) | 1.31–1.33 | 0.164–0.166 |

The baseline is **plausibly credible**. Its best-estimate errors are inside half the tolerance (1.5 %), but the drag error's uncertainty range (0.27–2.14 %) crosses that line. At matched domain size, the reference agrees with published results within about 1 %, the scatter between codes.

| Stage | Reference drag | What it revealed |
|---|---|---|
| 1 | 1.3133 ± 0.7 % | large-domain runs had not settled |
| 2 | 1.3046 ± 1.2 % | settled runs; St time error resolved |
| 3 | 1.290 ± 6.3 % | domain study does not converge cleanly |
| 4 | **1.3155 ± 0.94 %** | cause found: the outer mesh coarsens as the domain grows |

Getting a ±1 % reference took four stages and 13 extra simulations. Independent fact-checks of the draft reports pushed each step.

**A design flaw the project found in its own tool.** The "enlarge the domain" experiment keeps the growth rate of the cells, so the outer mesh gets coarser as the domain grows (the last cell grows from 1 to 4 diameters). Refining the outer mesh of the largest domain raised drag by 0.16 %. This skews every domain study the system runs. It makes the verifier more cautious (it overstates the domain effect), which is safe, but it cannot be accepted as a design. A domain experiment at constant outer resolution is the first thing to rebuild.

**A false rejection explained.** Every system rejects case J1, whose answers are within 1.1 % of the reference. The rejection comes from an extended mesh-quality check that fails on 48 stretched cells beside the outlet, 47 diameters behind the cylinder. That is a heuristic firing far from where the forces are produced.

## What we have learned so far

1. **Ordinary solver diagnostics are not verification.** OpenFOAM's own checks accepted 6 of the 7 clearly broken cases. The benchmark's flaws are not trivial for existing tools.
2. **Deterministic checks are safe but often silent on one run.** They have not accepted a single bad answer, but without extra simulations they decide only the set-up errors (steady solver, wrong viscosity, wrong reference).
3. **Targeted experiments expose numerical flaws.** One mesh refinement exposed A1, and one domain enlargement exposed B1, each at roughly the cost of the original run.
4. **Rigour is expensive and conservative.** Even the careful baseline cannot be formally ACCEPTed: its real drag error (about 1.2 %) is too close to the per-component limit (1.5 %) to be proven below it, after 5–6 extra simulations costing 7–8× the original run.
5. **An LLM reading the evidence alone is a strong baseline.** With no experiments and no checks it flagged all 7 broken cases from configuration "smells" such as a coarse mesh or an upwind scheme. But its ACCEPTs rest on no demonstrated accuracy. The attack cases, built to look clean, test whether it can be fooled.
6. **"Plausible but wrong" is often not wrong at 3 %.** Five of the twelve deliberately flawed set-ups (C1, E1, G1, G2, J1) land within 2 % of the reference. A useful verifier must quantify how wrong an answer is, not just spot bad practice.

## Still open

- **Finish the benchmark.** C-rule has decided 7 of 17 cases, C-llm 6 of 17 and the exhaustive protocol 2 of 17. The three attack cases have barely started (the inlet-shear attack is caught only by an extended check, and only as a set-up problem).
- **The pre-registered value tests.**
  - Does the adaptive system resolve cases the static verifier leaves open? So far yes: A1 and B1.
  - Does the LLM beat the fixed rule list on accuracy or cost? Mixed so far: cheaper on B1, dearer on the baseline.

  If the LLM adds no measurable value, the final report will say so.
- **Ablations.** Repeated LLM runs (how consistent are its choices?), an acceptance-review gate, and verifier extensions against the attacks.
- **Final report.** Answers to the 13 research questions: what it detects, what remains, false-acceptance rate, compute overhead, product potential, and what to build next and what not to build.

> **Caveats.**
> - One benchmark (laminar 2D cylinder) and 17 cases: rates are indicative, not statistically strong.
> - Several labels sit near the tolerance boundary, so two label views are reported: the pre-registered rule, and the rule with the reference uncertainty taken into account.
> - All numbers here are interim and will be updated when the runs complete.

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Repository: eklavyagoyal/socrates-cfd, branch claude/cfd-verification-prototype. Details: README.md, HANDOFF.md, reports/baseline_validation.md, docs/design.md (pre-registered protocol), docs/methodology.md (methods and the log of every correction).